The use and misuse of the Camouflaging Autistic Traits Questionnaire (CAT-Q) in autism research and clinical practice: Issues, considerations, and suggestions

Benjamin Ross Hannon*, Laura Hull*, Meng-Chuan Lai*, Iliana Magiati* & Will Mandy*

**Authors are listed in alphabetical order and all should be considered to have contributed equally to the conceptualisation, writing and revising of this work.*

Benjamin Ross Hannon - Cambridgeshire and Peterborough NHS Foundation Trust, Fulbourn, UK. ORCID ID https://orcid.org/0009-0001-2800-4121

Laura Hull - Department of Population Health Sciences, Bristol Medical School, University of Bristol, Bristol, UK. ORCID ID https://orcid.org/0000-0002-8289-2158

Meng-Chuan Lai - Campbell Family Mental Health Research Institute, Centre for Addiction and Mental Health, Toronto, Ontario, Canada; Department of Psychiatry, Temerty Faculty of Medicine, University of Toronto, Toronto, Ontario, Canada. ORCID ID https://orcid.org/0000-0002-9593-5508

Iliana Magiati - School of Psychological Science, The University of Western Australia, Australia. ORCID ID https://orcid.org/0000-0001-7261-7366

Will Mandy - Department of Clinical, Educational and Health Psychology, UCL, London, UK. ORCID ID https://orcid.org/0000-0002-3564-5808

Abstract

In this Perspective we summarise the evidence to date around the use of the Camouflaging Autistic Traits Questionnaire (CAT-Q) as a tool in research and clinical settings, and offer suggestions for its future use. We highlight that the CAT-Q is a useful measure of camouflaging/masking for research purposes, although questions remain regarding sociocultural variation in camouflaging experiences, and its specificity regarding autistic camouflaging compared to impression management more generally. With respect to clinical practice, we note that very little research has examined the use of the CAT-Q in clinical settings to date, and we encourage clinicians to be mindful of this limited evidence base and to refrain from using CAT-Q scores in diagnostic decision-making or as an intervention monitoring/outcome tool.

Background

Camouflaging, also known as masking, compensation, or a version of transactional impression management, involves hiding or changing autistic behaviours (whether consciously or not) as a social coping strategy to fit in with non-autistic social environments and norms[1-3]. Camouflaging

as a concept is not unique to autism: it has been considered akin to impression management in the general population[1, 4-6], it conceptually overlaps to some extent with social anxiety coping strategies[7, 8], and comparable experiences have been reported in other neurodivergent populations, such as in people with ADHD[9, 10], developmental language disorder[11], and tic disorders[12]. Further, other minoritised and marginalised groups, including LGBTQIA+ and culturally and linguistically diverse (CALD) people, may also conceal their sexuality, gender, linguistic, cultural or social background differences to protect themselves from stigma and discrimination (e.g. [13, 14]). However, there can be distinctive features of camouflaging as used by autistic people[15], such as the use of alternative cognitive processes to compensate for and learn social rules and strategies[16].

Measuring camouflaging in autism research and clinical practice:
The Camouflaging Autistic Traits Questionnaire (CAT-Q)

Several approaches to measuring camouflaging in autistic people have been developed and used to date, including self-report measures, observational approaches, and discrepancy approaches (e.g., exploring the discrepancies between self-reported/internal and observable autistic social and behavioural features; see [17] for a review; see also [18] for a critical discussion on the need for measurement precision). These approaches measure different aspects and facets of camouflaging, including the frequency of camouflaging behaviours used, the "effectiveness" of camouflaging, and/or the discrepancy between outwardly neuronormative behaviours and internal and self-perceived social differences and difficulties respectively.

Currently, the most widely used[19] self-report camouflaging measure in both research and clinical settings is the 25-item self-report Camouflaging Autistic Traits Questionnaire (CAT-Q[20]), which was co-developed by several of the authors of this Perspective; see also [21]. Drawing from autistic people's experiences[2], the CAT-Q was initially developed in English as a research tool to quantify the self-perceived frequency of use of camouflaging behaviours in autistic and non-autistic adults. It was validated in a large online study with autistic and non-autistic adults aged 16 years or older, and demonstrated good internal consistency and test-retest reliability[20].

In the initial validation of the CAT-Q, three factors were identified[17, 20]: Compensation (finding alternative approaches to neuronormative social behaviours, for example mimicking others' body language or studying and imitating social behaviours from films or books); Masking (hiding autistic characteristics, for example preventing oneself from stimming in specific situations); and Assimilation (strategies to fit in with others and avoid seeming different, such as 'performing' as neurotypical). Measurement invariance across assigned-sexes and autistic versus non-autistic groups has been demonstrated[20]. Beyond the initial validation study, evidence from a large, representative US general population sample also supports the three-factor structure[5]. However, further psychometric evaluation is warranted to determine whether three distinct but related factors is the optimal factor structure and preferable (or not) to the use of a single composite 'camouflaging' score. The CAT-Q has been translated into multiple languages, including Swedish, French, Dutch, Italian, Mandarin, Spanish, and Japanese so far[22-28], although findings regarding the psychometric properties of these translations are inconsistent, especially in relation to factor structure and measurement invariance.

Across several studies, higher CAT-Q self-reported ratings have been consistently associated with higher autistic traits and poorer mental health and wellbeing, usually with medium effect sizes (see reviews [18, 21, 29, 30]). Recent research has suggested that CAT-Q scores may positively correlate more strongly with measures of social anxiety than with measures of autistic characteristics, general anxiety or depression, pointing towards complex interrelations among camouflaging and social anxiety[1, 5, 7, 8, 31]. A recent study found although overall CAT-Q scores did not change following a social anxiety intervention for autistic adults, the social anxiety and camouflaging change scores were significantly correlated, suggesting that these are distinct but related constructs[32].

In younger age groups, the CAT-Q has recently been psychometrically evaluated with small groups of autistic and non-autistic adolescents in Sweden[27] and in Taiwan[26], with good to excellent internal consistency, good test-retest reliability, and good convergent validity with measures of mental health, similar to findings in the adult literature. A parent-report version has also been developed for, and was used with, 12-18 year-old young people in the UK and Taiwan, although this has not yet been thoroughly psychometrically evaluated[17, 26].

The use of the CAT-Q in autism research

There are several considerations regarding the use of the CAT-Q in research, as noted previously[17, 18, 33]. One key concern is its potential confounding with social anxiety, raising questions as to whether the CAT-Q measures camouflaging behaviours per se, or broader social anxiety related behaviours[34]. For instance, the CAT-Q Masking and Compensation subscales were more strongly associated with social anxiety symptoms (r=.25 and .45 respectively) than with autistic traits (r=.04 and .29 respectively)[7]. The CAT-Q Assimilation subscale items have been shown to load together with both social autistic traits and social anxiety items in factor analyses[8], and a network analysis across CAT-Q and social anxiety items shows a small number of social anxiety items being grouped together with the CAT-Q Assimilation item community[31], suggesting some plausible conceptual and measurement overlap. Prospective follow-up and longitudinal studies are required to understand the extent to which the association between camouflaging and social anxiety reflects a causal directional relationship (e.g., individuals experiencing social anxiety might be more likely to camouflage) and/or whether bi-directional relationships exist (e.g., whether camouflaging and social anxiety reinforce each other over time).

Moreover, the CAT-Q was developed to capture camouflaging behaviours based on autistic people's experiences - it was not designed or intended as a screening measure of autistic traits. Nonetheless, some studies report statistically significant, but mostly small to medium effect size, camouflaging group differences (with autistic teens or adults reporting somewhat more camouflaging behaviours than non-autistic participants; e.g. [23, 26, 35]), while other studies report no significant group mean differences between autistic and non-autistic individuals in total or subscale CAT-Q scores (e.g., the original CAT-Q Masking subscale[7, 20, 36]; and in some translated versions of the CAT-Q[27, 28]). Receiver Operating Characteristic (ROC) curve analyses in a recent study[27] further show that the Swedish CAT-Q did not have adequate sensitivity and

specificity in distinguishing between autistic and non-autistic participants, with sensitivity not exceeding .51-.58. Although examining group differences in research can provide insights into group differences in average patterns of camouflaging, such analyses can easily be misinterpreted – particularly outside academic and research settings – as suggesting that the CAT-Q may serve as a "screening" or "diagnostic" discriminator between autistic and non-autistic people. This is not the case. There is currently no theoretical or empirical justification for using the CAT-Q as a proxy of autistic traits to differentiate between autistic and non-autistic individuals.

In addition, there may be age, cohort, gender and/or sociocultural effects in camouflaging behaviours. Although autistic adults may maintain similar CAT-Q scores over time, at least based on a recent study with two annual follow-ups[37], it has been suggested that camouflaging behaviours tend to decrease with age in non-autistic adults[1, 27]. Gender influences have also been reported, with some, but not all, studies indicating that autistic women tend to report more camouflaging behaviours than autistic men[21, 38], and gender diverse autistic adults report higher Compensation subscale scores than cisgender autistic adults[38]. Sociocultural effects in camouflaging have also been noted[3], with studies in several different countries (e.g. [25, 28, 39]) showing at least some differences in CAT-Q measurement properties and associations with mental health in different ethnic and sociocultural contexts. However, age, cohort, gender or sociocultural effects and influences are often not considered in research to date largely conducted with adult participants spanning in age from their early 20s to late 70s.

More extensive examinations of the CAT-Q are therefore needed to clarify the most appropriate research contexts and applications for its use, and to inform the development of alternative methods for measuring camouflaging across ages, abilities, and sociocultural contexts. Notably, the various factor structures found in different language versions of the CAT-Q highlight the limitation and caution of a direct comparison of scores across sociocultural contexts, and the need to develop complementary methods to capture camouflaging in local sociocultural contexts with enhanced validity.

The use of the CAT-Q in clinical settings

Although initially developed as a research tool to measure the frequency of use of camouflaging behaviours, the CAT-Q is increasingly being incorporated into clinical settings. However there is currently little to no empirical evidence examining whether, how, or for/with whom the CAT-Q is used in clinical practice, nor exploring and clarifying the specific clinical purposes it may serve.

This Perspective draws from our research and clinical experiences, informal discussions with some clinicians in the UK, US, Canada, Taiwan and Australia, as well as from reviewing public online resources that describe or recommend the use of the CAT-Q for clinical purposes. Currently, it seems that clinicians tend to use the CAT-Q within broader mental health or autism-specific clinical settings in two main ways:

(1) as part of screening, triage, and/or autism and related neurodevelopmental diagnostic assessments; and/or

(2) to facilitate information gathering, formulation and support planning as part of wellbeing, mental health, post-identification or related psychosocial care services for autistic people.

Below we discuss both potentially problematic or concerning clinical use and likely helpful use of the CAT-Q in relation to (1) and (2).

1. (Mis)Using the CAT-Q in screening, triage, and/or diagnostic assessment

Assessment of autistic characteristics

In many services (for example, in several National Health Service UK trusts), adults referred for an autism assessment may be screened or triaged to determine whether they should be referred for a comprehensive autism diagnostic assessment. Based on our readings, searches across different websites, and discussions with clinicians and researchers, it seems that the CAT-Q is sometimes incorporated in screening, triage, or diagnostic assessments as a tool to help guide decisions around who should be referred for a comprehensive autism diagnostic assessment or who may eventually be diagnosed as autistic. In some services, the CAT-Q has been incorporated or suggested as a screen alongside other established and commonly used screening measures of autistic characteristics, such as the Autism Spectrum Quotient[40] or the Social Responsiveness Scale-2 (SRS-2[41]). Examples can be found in clinical recommendation papers[42], and on the websites of certain private autism assessment services and autism information organisations. In some of these settings, we have read or heard that higher CAT-Q scores are being interpreted as an indicator of specific 'internalised' autism presentations (explaining why autistic characteristics are less observable), or as an explanation for why an individual eventually diagnosed as autistic does not meet the algorithm or criteria thresholds using tools that measure diagnostic, core features of autism. For instance, when other validated observational diagnostic tools such as the Autism Diagnostic Observation Schedule (ADOS-2[43]) do not reach the cut-off for an 'autism' or 'autism spectrum' classification, it is sometimes suggested that the CAT-Q can be used to identify autistic individuals who would otherwise meet diagnostic thresholds, if not for their ability to mask their autistic behaviours.

As the CAT-Q was not designed nor validated to measure autistic characteristics as set out in diagnostic criteria, **CAT-Q scores cannot, and should not, be used to suggest 'above' or 'below' cut-off reports or autistic vs. non-autistic group membership**. In other words, the CAT-Q is not a screening tool for autism, and autism screening measures used should reflect measurement of autistic traits regardless of whether or not these may be camouflaged[44]. Importantly, **there are no threshold or cutoff scores on the CAT-Q** regarding the extent of camouflaging in different groups or the presence of autistic characteristics. Evidence suggests that not all autistic people camouflage, that autistic people camouflage to different degrees in different contexts, that camouflaging strategies vary widely, and that non-autistic people as well as people with other neurodivergence or other differences may also camouflage at high levels[15, 20, 45]. Therefore, **the CAT-Q should not be treated or used as equivalent to measures of autistic characteristics for the purposes of screening, triage, or diagnosis.**

At the same time, **incorporating the CAT-Q as part of one's comprehensive assessment toolbox of multiple measures, informants and approaches, may aid the assessment** by providing additional information to inform expert case discussion and decision making for clinical formulation. Specifically, using the CAT-Q may provide insights into the extent to which the individual, as they were growing up, developed masking or compensatory strategies to understand and behave in different social situations. Such information may be particularly relevant to clinical information needed to establish autism diagnostic criterion C of the DSM-5-TR[46], namely that autistic characteristics and features "*must be present in the early developmental period (but may not become fully manifest until social demands exceed limited capacities or may be masked by learned strategies in later* life)"[47]. Inviting autistic adults to complete the CAT-Q, and then following up with more in-depth developmental inquiries could provide useful information regarding the extent to which an autistic person may be hiding or changing autistic social or behavioural characteristics in social situations currently, and the extent to which they may have been carefully 'studying' or 'learning' social rules and conventions over time to navigate complex social environments, when most non-autistic people tend to acquire such learnings more intuitively[16].

To determine the extent to which a person's current and/or historical features meet diagnostic criteria for autism, clinicians should conduct a comprehensive developmental interview to collect evidence of autistic characteristics and experiences in childhood, adolescence, and adulthood. They also need to gather current information on social and behavioural autistic features via self- and informant-reported measures of autistic characteristics, in-depth interviews and through clinicians observing the person's presentation and behaviours, ideally on more than one occasion. The clinicians should integrate this developmental interview with information from the CAT-Q amongst other measures and follow-up discussions of the responses to develop a fuller picture of the individual's social experiences and ways of navigating social environments over time and across different contexts (for instance, at work versus at home). Such information needs to be interpreted and understood in a developmental and biopsychosocial framework to support a diagnostic clinical formulation[44]. However, we emphasize again that although useful as an information gathering tool (together with comprehensive information gathering on autistic characteristics across the person's lifespan), **the CAT-Q is not a screening tool for autism, as there is no threshold or cut-off score at which an individual might be considered to 'be camouflaging' or 'be camouflaging because they are autistic'**.

Camouflaging is not a universal autistic characteristic

As a self-report measure, the CAT-Q ratings represent a continuous distribution of behaviours across the entire population, including those who do and do not meet diagnostic criteria for autism[5]. There is evidence of heightened CAT-Q scores amongst individuals with other conditions such as ADHD, social anxiety, and depression[7, 10], and in other minoritised groups[48], as well as a range of endorsement of CAT-Q items in the general population, suggesting camouflaging behaviours are not specific or unique to autism. On the other hand, some autistic people may engage in very little or no camouflaging. Thus, when the CAT-Q is included as one

tool in a comprehensive assessment, **the CAT-Q score alone should not be the basis for confirming or disconfirming an autism diagnosis.**

Importantly, camouflaging is not a diagnostic criterion of autism within either DSM-5-TR or ICD-11, although there is reference to masking and compensation when explaining heterogeneity in the developmental course and diagnostic features of some autistic people. At this stage, there is little evidence to suggest camouflaging should be included in the diagnostic criteria for an autism diagnosis, while there is considerable evidence to show that camouflaging is not specific nor unique to autistic people, as many other neurodivergent or minoritised people camouflage[49]. Some recent measures of autistic traits in adults have included items, and a subscale, relating to camouflaging (e.g. the Concise Autism Trait Inventory or CATI[50]), which can provide useful information, however we caution against the assumption that all autistic people camouflage, or that camouflaging should be used as evidence of autism when observational or other gathered historical and current information is not consistent with the diagnostic features of autism. It is heartening to see that in a recent survey of clinicians, only 8% agreed that camouflaging should be considered a 'sign' of autism[51]. We do argue that there are some cases where historical evidence of autistic characteristics across development (via self-report, accounts from other informants, and clinical information gathering and observations) can be combined with evidence of camouflaging (and, most importantly, its development over time) to inform a diagnostic formulation of autism[47], but that camouflaging is only one piece of data informing the clinical diagnostic decision.

To date, some sex and gender differences in levels of camouflaging have been reported in autistic and non-autistic people[19], suggesting that clarifying camouflaging, especially learned strategies and approaches to understand and manage social situations, may be particularly useful when understanding the experiences of autistic women and those assigned female at birth[38, 52], in both assessment and therapeutic contexts. However, autistic people of all sexes and genders report varying levels of camouflaging, so **it is not appropriate to consider camouflaging solely as a sex- or gender-specific phenomenon.**

Much of the existing research using the CAT-Q has drawn on online-recruited autistic adults self-reporting their diagnosis or self-identifying as autistic. These studies, most of which include autistic or possibly autistic adults (often majority women) with high rates of co-occurring mental health or other neurodevelopmental conditions, have made important contributions to our understanding of camouflaging. Nevertheless, online samples may underrepresent autistic people often seen in clinical services, for instance those with intellectual disability, those with little spoken language, or those with complex physical, mental, or other neurodevelopmental conditions. As such, the extent to which the broader population camouflages, and whether the CAT-Q is an appropriate tool to measure camouflaging for them, is unclear. Further research is needed to examine whether and how camouflaging is used by autistic people in studies within clinical settings, and the utility of the CAT-Q in clinical settings. Research should also explore if additional or alternative methods of measuring camouflaging would be beneficial.

Clinicians may find it helpful to use CAT-Q ratings, alongside other information regarding social coping experiences and strategies throughout one's life, to inform detailed conversations with the individual regarding their experiences and to help clinicians make sense of multifaceted assessment information for the overall diagnostic formulation, following best practice guidelines[44, 53]. For instance, a clinician might identify items or subscales within the CAT-Q which the participant highly endorsed and use this as a starting point in the assessment interview to discuss their social coping experiences in different situations, and how these evolve over time. This may contribute to exploring dilemmas or contradictions in the assessment data, for instance if there is discrepancy between observed and self-reported autistic characteristics or between current autism presentation and developmental/historical information. For example, an individual may endorse autistic traits above established screening cut-off scores for possible autism when completing self-report measures of autistic traits, and when that same individual provides extensive clinical information relating to social, communication and behavioural autistic characteristics. If that individual also endorses high use of certain camouflaging behaviours in the CAT-Q, but few current autistic behaviours are observed by the clinician in an ADOS-2 or other semi-structured assessment session, then – and assuming the clinician coded the assessment tool reliably and in a valid way – this discrepancy needs to be investigated further alongside in-depth clinical discussions to aid the diagnostic formulation.

There is very limited research regarding whether and how camouflaging is considered during real diagnostic assessments. A recent survey of autistic adults and clinicians with experience in adult autism diagnostic assessment revealed that over 80% of each of these groups felt that camouflaging influences the autism assessment process[51]. Clinicians were not asked if they had any concerns regarding the integration of camouflaging to a comprehensive diagnostic assessment, although one theme identified was that camouflaging ('masking') affected clinicians' interpretation of diagnostic tools, suggesting some impact. Only 19% of autistic participants thought that camouflaging had been considered during their own assessment, compared to 73% of clinician participants who reported taking camouflaging into account during assessments. While the authors acknowledge that the clinicians included in this study may not represent those performing assessments across the general population, it nevertheless suggests that there is a need to better communicate with people being assessed about whether and how camouflaging is being considered during an assessment. This was reflected in the survey, where 51% of clinicians called for better and more validated tools to assess camouflaging[51]. The CAT-Q, and future methods of measuring camouflaging, could offer one approach as a starting point for further discussion of how camouflaging may impact the individual.

2. Use of the CAT-Q to inform formulation and understanding of autistic people's mental health and wellbeing in clinical settings

In several cross-sectional qualitative and quantitative studies, higher camouflaging, as measured by the CAT-Q, has been consistently associated with greater self-reported generalized anxiety, depression, social anxiety and suicidality[7, 21, 30, 54]; for this reason, clinicians

in assessment and therapeutic settings also use the CAT-Q when they wish to explore with their autistic clients how camouflaging may be related to their mental health, wellbeing, and/or identity. Camouflaging may also be useful to explore when considering differential and concurrent diagnoses or when identifying safety behaviours associated with social anxiety. One follow-up study found no overall relationship between camouflaging and mental health in autistic adults across a two-year period, but that those who camouflaged more at baseline may have slightly increased mental health problems[55]. Except for this study, whilst cross-sectional associations between camouflaging and mental health have been consistently reported, the directionality and temporal sequence of this relationship remain largely unclear.

Within clinical settings, the CAT-Q is also increasingly being used as a measure to support discussion and understanding of autistic clients' psychosocial stressors and experiences motivating camouflaging, such as stressful life events, discrimination, and stigma (see [30, 56] for reviews), and/or the impact that these may have on the individual (see also [51]). For instance, the CAT-Q may be used to quantify and discuss camouflaging strategies that an individual may find hard to identify on their own and to prompt more in-depth discussion of specific strategies, when and how they are used, and the consequences. This information can be useful for diagnostic formulation of mental health issues and as part of therapeutic formulation of an autistic client's wellbeing.

The CAT-Q may also be helpful when considering post-identification or post-diagnostic supports. If a person scores highly[i] on the overall CAT-Q or on subscales, it may suggest a high level of stigma, shame or other psychosocial pressures and motivations around their autism-related experiences[3, 30], which could be explored, understood, and supported. As such, CAT-Q data could point to a need for, and assist with, further exploration of the person's social and psychological experiences: for instance, supporting the autistic person to reframe their earlier negative social experiences; consider where, with whom and how they can be their authentic selves; and develop a more positive autistic identity and optimal impression management that benefits, rather than exhausts or distresses them[15]. Using the CAT-Q as an initial exploration of psychosocial experiences could contribute to the development and implementation of interventions on their social environment, for example, changing attitudes of those around the autistic person[57], and/or individual work with the autistic person towards processing and making sense of their social differences and ways of coping towards self-acceptance and self-care[58].

As autistic individuals tend to have disproportionately high rates of mental health issues[59] and camouflaging may be one of several contributing factors[60], the potential relationship between camouflaging and the person's wellbeing may also be explored as part of therapeutic or other psychosocial support work. **However, we recommend that clinicians take an individualised approach to camouflaging when supporting autistic adults.** For some individuals, camouflaging may be a compelled, unwanted and harmful response to negative treatment by others, for which support to safely 'drop the mask' towards disclosure and acceptance may help. For others, camouflaging, as part of the ubiquitous impression management experiences of human beings, may be a pragmatic, protective, functional and necessary set of strategies to get through complex social situations, succeed in education or work, maintain relationships, and

reduce harm[30]. In discussions between one of the authors and two of their research team's autistic advisors, the advisors have highlighted that camouflaging should not be something clinicians necessarily encourage autistic clients to fully 'drop'; rather, they emphasised the need for an individualised, nuanced and contextual understanding of camouflaging – its benefits, the needs for it, and its costs and alternatives (see also [58]). Clinicians should avoid making assumptions about the motivations for, extent of, and impact of camouflaging for any individual. Rather, they should collaborate to recognise that the goal of therapeutic support is not to eliminate camouflaging, but to support autistic people to reflect on when, where and with whom they may be able to express themselves more authentically – or when they might choose disclosure and request reasonable adjustments instead of camouflaging.

Relatedly, we also suggest that clinicians collaboratively identify meaningful individualised therapeutic goals and outcomes to monitor with their autistic clients, rather than relying solely on changes in camouflaging as an indicator of intervention/support impact. At present, there is very little research exploring change or continuity in camouflaging across meaningful timepoints. To our knowledge, two published studies have assessed change, both suggesting that CAT-Q scores remain broadly stable over 1-2 years, with only small declines in camouflaging at the group level[37, 55]. Further investigation of how camouflaging evolves across the lifespan – and whether the CAT-Q is sensitive to change – would help researchers and clinicians contextualise camouflaging stability or change over time, or following support provision. Because the CAT-Q measures camouflaging frequency, changes in scores are unlikely to capture potential changes in other aspects of camouflaging, such as its intentionality, effort, goal-attainment or impact. Therefore, **changes in CAT-Q scores should not currently be interpreted as a standalone measure of clinical outcomes**.

Use of the CAT-Q in child and adolescent clinical settings

As the CAT-Q was developed and validated for use with adults initially, we have so far focused on using the CAT-Q in adult settings. While there are approaches to measuring camouflaging which may be suitable for use with children and adolescents, such as the discrepancy approaches[61, 62], these may be more time-consuming and/or have limited validation in research settings[17]. A parent-reported CAT-Q has been developed[17] which asks parents or carers to report on camouflaging behaviours observed or reported by their child. Although the parent-reported CAT-Q correlates positively with self-reported CAT-Q scores with a medium effect size (r~0.4) in *adolescents* in the UK[17] and Taiwan[26], it has not undergone thorough psychometric evaluation, and its use in children and in people with different cognitive abilities is essentially unknown. If older children and adolescents, and/or their caregivers, can report on camouflaging strategies, the CAT-Q may be a useful tool to gather information as we outlined earlier. However, we again emphasise that **camouflaging presence or amount, especially the CAT-Q total and subscale scores, should not be used as a metric of autism-related traits for screening or diagnosing autism in children or adolescents**.

A key challenge in identifying camouflaging in paediatric clinical settings is that little is known about the presentation, nature, and development of camouflaging earlier in life. Although some studies report that autistic people are able to identify and describe their camouflaging from

middle childhood onwards[63, 64], it is not yet clear (a) whether, and to what extent, these behaviours and experiences differ from the impression management strategies described by typically developing children and adolescents[65] or experimentally inferred in even earlier stages of life[66]; (b) when such behaviours and experiences begin; and (c) whether or how they may be uniquely experienced in autistic children and adolescents. As such, at present we recommend caution when using any tool, including the CAT-Q, to measure camouflaging in children and adolescents, in research as well as clinical settings.

Suggestions for use of the CAT-Q in clinical settings based on current evidence

To conclude, we summarise some of our reflections on the ways in which the CAT-Q may be used to support clinical practice, based on the empirical evidence to date and our research, clinical and lived experiences. We note that several of these recommendations were also suggested by clinicians and autistic individuals in a recent study exploring how camouflaging influences adult autism assessments[51].

Given insufficient evidence to date, the CAT-Q should not be used to screen for autism, nor should cut-off scores be applied to classify individuals as 'high' or 'low' camouflagers, to infer 'high or low-masking autism', or to determine autistic status. Similarly, the CAT-Q should not be used as the primary or sole measure to evaluate clinical interventions or track therapeutic outcomes over time. In brief, **the CAT-Q is not a screening or diagnostic tool, and its total or subscale scores should not be used to determine diagnostic or intervention decisions**. Rather, information collected from the CAT-Q may serve as a starting point for a detailed developmental exploration of whether and how a person may have learned to modify autistic behavioural characteristics in social situations over time. Such detailed information, collected through clinical interview, the CAT-Q, and other assessments, may then contribute to a diagnostic formulation.

Therefore, the CAT-Q as a measure of self-reported aspects of camouflaging behaviours may be used currently in adult assessment settings:

1. To contribute to multi-method comprehensive assessments as part of a range of standardised measures.
2. To deepen understanding of the autistic person's lived experiences across development, specifically through (a) exploring their social coping strategies and experiences growing up, including stigmatisation, discrimination and other psychosocial experiences; (b) facilitating discussions aiming to foster a stronger autistic identity and adaptive impression management strategies that can support social coping and wellbeing; and (c) informing clinical psychosocial formulation of wellbeing, including collaborative exploration of strategies to enhance wellbeing, which may involve re-evaluating their use of camouflaging.

Future directions

Overall, the limited evidence to date suggests that measuring camouflaging could potentially be useful as part of exploring and understanding autistic adults' psychosocial experiences,

particularly when considering their mental health and psychosocial wellbeing. However, much more research is needed to determine the usefulness of the CAT-Q in clinical settings, from the perspectives of both clients and clinicians. Understanding how camouflaging manifests and is experienced in young people and people with higher support needs remains a major research gap. Other (existing and novel) methods of measuring camouflaging may be more useful within mental health settings, as the CAT-Q only measures the frequency of use of a limited range of camouflaging strategies. Tools to capture an individual's motivations for camouflaging, the effort it takes, their capability to do so, and the impacts (both positive and negative) it has will be worth developing and evaluating.

More information is also needed regarding the CAT-Q's utility as an information gathering and/or assessment tool with autistic people and with people with other conditions where social coping and masking experiences may also be salient, such as social anxiety, depression, ADHD, tic disorders, language disorders, and learning disabilities.

These important gaps suggest the following questions could be prioritised in future research:

1. How does camouflaging develop over time and in whom? When does it begin, why, and how does it change at different developmental stages?
2. How can information about camouflaging, including information collected by the CAT-Q, other existing or novel measurement approaches, and clinical interview, be helpful and effective in assisting clinical decision-making in relation to autism diagnosis, differential diagnosis, and mental health care provision?
3. What are the impacts of camouflaging over time, for different individuals? Prospective longitudinal research is needed to establish directional relationships.
4. Are there other ways we can improve current measures, including the CAT-Q, or develop new ways to measure camouflaging which capture its different facets, including the effort, effectiveness and consequences of camouflaging for the individual over and above the frequency of use of camouflaging behaviours?
5. Can the CAT-Q or other camouflaging measures be used to assess psychosocial and wellbeing clinical outcomes following therapeutic support?

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[i] We note here that 'highly' is a subjective term, and that there is no established cutoff or threshold score to represent 'high' camouflaging on the CAT-Q, due to the extreme variation in camouflaging between different individuals and the fact that the CAT-Q does not capture all expressions of camouflaging. In this context, we would encourage clinicians to use their own judgement when interpreting CAT-Q scores, for instance considering whether an individual's scores are higher on certain subscales compared to others, or if there are items or clusters of items which the individual 'strongly agrees' with.